Product Requirements Document (PRD) — Queueless

Version 1.0 (MVP) · Web Application Specification

Field	Details
Product Name	Queueless
Platform	Web Application (Mobile-first responsive layout)
Version	v1.0 (MVP)

1. Project Overview

- **App Name:** Queueless
- **One-Line Description:** A platform that helps users shop smarter by comparing prices across nearby stores, checking product availability, and avoiding unnecessary queues and shopping trips.

2. Problem Statement & Goals

- **What problem does this app solve?** Consumers waste time and money visiting physical stores only to find items are out of stock, overpriced, or checkout queues are excessively long.
- **What are the main objectives of this project?** To provide real-time price comparisons, cross-store basket cost transparency, item availability tracking, and user-reported store queue metrics so shoppers can plan optimal, efficient shopping trips.

3. Target Audience

- **Who will use this app?** Budget-conscious shoppers, families managing household grocery expenses, university and college students, young professionals, and everyday consumers looking to reduce travel costs and checkout wait times.

4. Key Features (Scope)

- **Feature 1 (User Authentication):** Secure registration and login for Shoppers and participating Retailers.
- **Feature 2 (Location-Based Store Discovery):** Ability to filter and view nearby physical retail outlets within a specified suburb or distance.
- **Feature 3 (Product Search & Comparison):** Search product catalog and view localized alternative store prices ranked from lowest to highest.

- **Feature 4 (Stock Availability Indicators):** Clear display states highlighting item inventory status (`In Stock` , `Limited Stock` , `Out of Stock` , `Unknown`) with price update timestamps.
- **Feature 5 (Shopping Basket Cost Aggregator):** Virtual multi-item basket compilation that aggregates and compares total checkout costs across different store branches to recommend the cheapest destination.
- **Feature 6 (Crowd-Sourced Queue Level Reporter):** Simplified community submission grid allowing users to report physical line wait times at a store branch checkout.
- **Feature 7 (Management Dashboards):** An Administrator view to manage retailers, stores, and products, alongside a Retailer panel for catalog uploads and inventory syncs.

4.4 Business Rules

ID	Rule	Used By
BR2	**Report Throttling Lockout:** A user is restricted to a maximum of 5 queue updates and 3 price changes per store branch within any single 60-minute window to avoid data spamming.	F3, S5
BR3	**Basket Consolidation Rule:** If a particular store branch does not carry a specific item present within a user’s current multi-item basket, the overall total screen must flag that store with "Missing 1 Item" and calculate aggregate sums based strictly on the available inventory set.	F2, S4
BR4	**Queue Decay Weighting:** The displayed live queue level indicator is calculated using a time-weighted average where user reports submitted within the last 15 minutes hold 80% weight, decreasing linearly to 0% weight over 2 hours.	F1, F3, S3

4.5 Notifications

Trigger	Recipient	Channel	Exact Text	Tapping Opens
User crowd-sourced price/queue report is verified	Shopper	In-app notification	"Thank you! Your update for Shoprite Hatfield has been verified. +10 points added."	S6 Savings Analytics Dashboard
Total cost of items in saved basket changes by >10%	Shopper	Email	"Price drop alert: Your active grocery basket is now R45 cheaper at Pick n Pay Hatfield!"	S4 Shopping Basket
Retailer catalog upload completes successfully	Retailer	Email	"Catalog sync complete: [Number] products updated successfully."	Database dashboard

5. User Flow, Tech Stack & Accounts

5.1 User Flow

1. A user lands on the Home Screen and sets their location or suburb filter.
2. They perform a product search or select an active shopping list basket.
3. The app presents the ranked price and total cost options across nearby stores, accompanied by queue indicators.
4. The user selects their destination, visits the store, and optionally logs a quick queue or price report check from the store detail view.

5.2 Technologies Used

- **Frontend:** React + Vite (Web view layouts optimized for mobile browser viewports)
- **Styling:** Tailwind CSS (Modular, light utility framework classes)
- **Backend & API:** Node.js + Express / Supabase Client Layer
- **Database:** PostgreSQL (Relational index pairing for products, listings, and user records)
- **Log-in & Storage:** Supabase Auth and Storage Buckets

5.3 Accounts and Sign-in

Decision	Specification
Sign-up methods	Email + password only
Who may sign up	Anyone (open public registration for shoppers; retailer accounts approved by system admin)
Minimum age	16 (Sign Up has the mandatory checkbox "I am 16 or older")
Email or phone verification	Required via email link. Accounts remain restricted to read-only guest status before verification.
Password rules	Minimum 8 characters, including at least 1 uppercase letter and 1 number. Show/hide toggle icon included.
Forgot password	"Forgot password?" link sends a secure reset link via email, valid for exactly 1 hour.
Failed log-in attempts	Account is temporarily locked for 15 minutes after 5 consecutive incorrect password entries.
Stay logged in	Session remains active for 30 days before requiring a manual password sign-in re-check.
Log out	Accessible from S6 Profile menu tray; log out is immediate with a single tap confirmation pop-up.

Delete account	Triggered in user settings; permanently deletes profile, email logs, and saved baskets within 7 days.
Change email or phone	Allowed via settings; requires verification code validation sent to the existing verified email address.

6. User Stories, Acceptance Criteria & Edge Cases

6.1 Stories

Story	Feature	User Story	Acceptance Criteria (with Test IDs)
US1	F1	As a Shopper, I want to search for a specific grocery product by its name so that I can see which local retail branch has the lowest price.	**T-US1-1:** Given I am on the S1 Search landing page, when I type "White Sugar 2kg" into the lookup bar and tap "Search", then I am redirected to S2 containing a list of matched item results ordered lowest price first. **T-US1-2 (unhappy):** Given the search text yields zero system index matches, when the submission executes, then the view container displays the message "No products found matching your search term. Please try another product name."
US2	F2	As a Shopper, I want to add three regular items to a shopping basket so that I can see the aggregate cost comparison across multiple competing stores.	**T-US2-1:** Given I am viewing product results on S2, when I click "Add to Basket" on consecutive rows, then my persistent header badge increments dynamically, and navigating to S4 displays grouped row items with accurate calculated totals. **T-US2-2 (unhappy):** Given my network connection disconnects while appending an item row, when the button is triggered, then an overlay message states "Action failed. Please verify your internet connection and try again."
US3	F3	As a logged-in Shopper at a physical store check-out, I want to submit a quick update on current line congestion so that other shoppers can skip long queues.	**T-US3-1:** Given I am on the S3 Store Details page, when I tap the "Report Queue" button, select the "Long Line (30+ Mins Wait)" grid option, and hit "Confirm", then the master record updates and a toast confirmation stating "Queue report submitted. Thank you for your contribution!" displays. **T-US3-2 (unhappy):** Given I attempt to submit a queue report for the exact same store branch 5 minutes after a previous entry, when I click confirm, then the form layout blocks entry with a red label reading "Submission restricted. You can report queue states once every 15 minutes."

6.2 Edge Cases

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